

ALL



2. Which interrupt is defined for system calls in Linux?

Which interrupt is defined for system calls in Linux?

Pick **ONE** option

☐ 0x10

☐ 0x60

☐ 0x80

☐ 0x40

Clear Selection

3. Operating system maintains the information about a process

Operating system maintains the information about a process

Pick **ONE** option

3. Operating system maintains the information about a process

Operating system maintains the information about a process in which of the following?

Pick **ONE** option

☐ Heap

☐ Process Control Block

☐ Stack

☐ TLB Cache

Clear Selection

4. Why is a virtual memory address called 'virtual'?

Why is a virtual memory address called 'virtual'?

Pick **ONE** option

TLB Cache

Clear Selection

4. Why is a virtual memory address called 'virtual'?

Why is a virtual memory address called 'virtual'?

Pick **ONE** option



This memory address does not map to actual memory cell that corresponds to address number.



This memory is the same as physical memory.



This memory cannot be accessed by applications at run time.



This memory address is always present even after a system is restarted.

Clear Selection



Saved!

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7. Consider the below java method. What's the output of
`temp = data[low]; data[low] = data[high]; data[high] = temp`

Consider the below java method. What's the output of this method?
`public static void function_1(int [] data, int low, int high) {
 if (low < high) {
 int temp = data[low];
 data[low] = data[high];
 data[high] = temp;
 function_1(data, low +1, high -1);
 }
}`

Pick **ONE** option

- ☐ The sum of all the elements of the array
- ☐ The product of all the elements in the array.
- ☐ The last element of the array.
- ☒ Reverses the contents of the array.

Clear Selection

 Saved!

9. Which of the following is a key-value store NoSQL database system?

Which of the following is a key-value store NoSQL database system?

Pick **ONE** option

☐ HBase

☐ MongoDB

☒ Redis

☐ Accumulo

Clear Selection

 Saved!

10. What is the latest long-term release of Oracle Data

What is the latest long-term release of Oracle Database?

11. Which of the following is not a property of Relatio

Which of the following is not a property of Relational Database?

Pick **ONE** option

☐ Atomicity

☐ Consistency

☐ Isolation

☒ Dependency

Clear Selection

 Saved!

12. What does CAP theorem describe with respect to Sy

What does CAP theorem describe with respect to System Design ?

Pick **ONE** option

17. Which is the HTTP request method used to retrieve the metadata associated with the resource's state?

Which is the HTTP request method used to retrieve the metadata associated with the resource's state?

Pick ONE option

☐ GET

☐ PUT

☐ HEAD

☐ POST

Clear Selection

17

18

19

20

19. What is the purpose of ARP (Address Resolution Protocol) in computer net

What is the purpose of ARP (Address Resolution Protocol) in computer networking?

Pick **ONE** option

- ☒ To convert IP addresses to MAC addresses
- ☐ To convert domain names to IP addresses
- ☐ To establish a secure connection between client and server
- ☐ To route packets between networks

Clear Selection

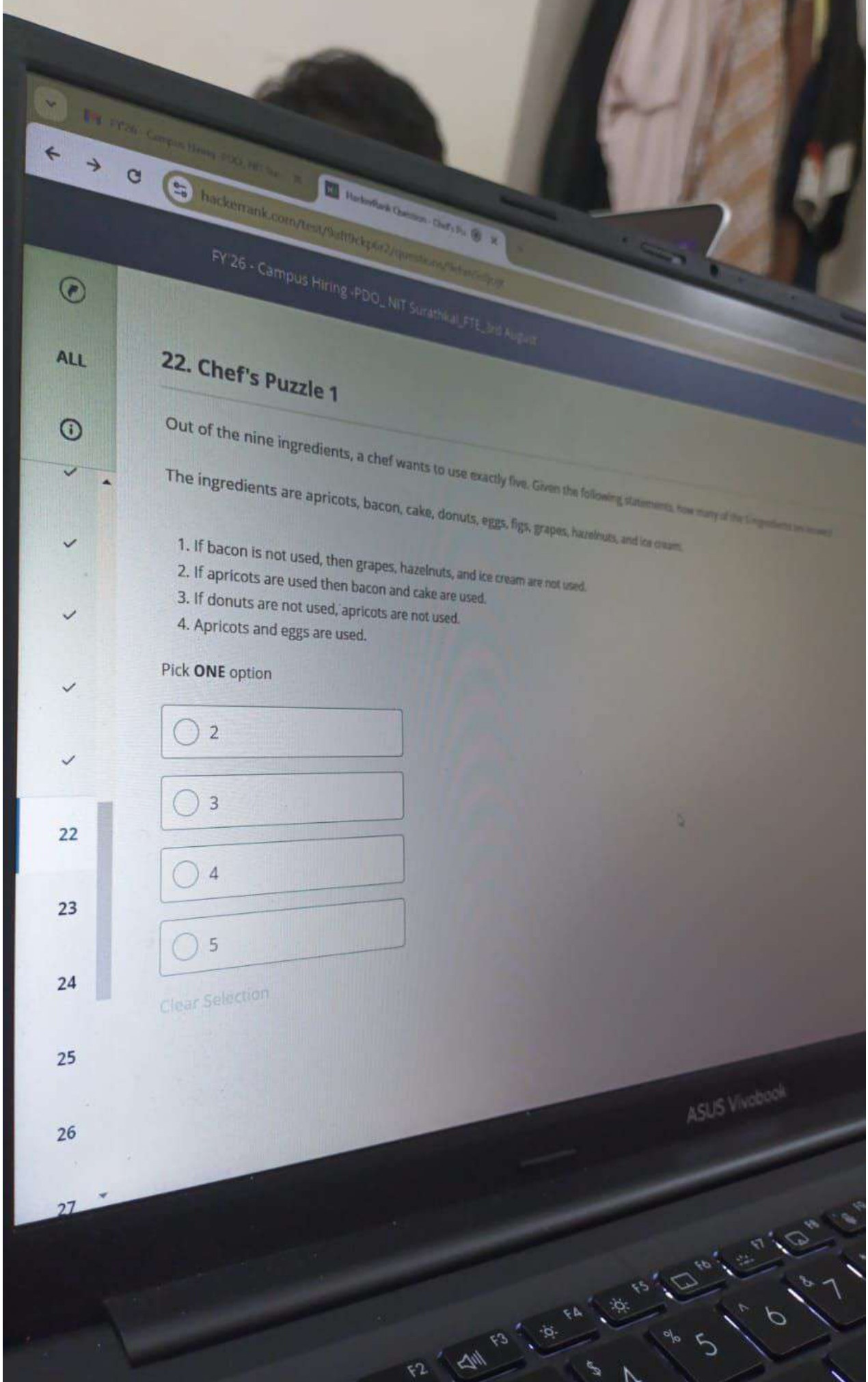
✓ Saved!

20. In what way the Symmetry Sparse Matrix can be stored efficiently?

In what way the Symmetry Sparse Matrix can be stored efficiently?

Pick **ONE** option

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22

23

24

25

26

27

22. Chef's Puzzle 1

Out of the nine ingredients, a chef wants to use exactly five. Given the following statements, how many of the ingredients are used?

The ingredients are apricots, bacon, cake, donuts, eggs, figs, grapes, hazelnuts, and ice cream.

1. If bacon is not used, then grapes, hazelnuts, and ice cream are not used.
2. If apricots are used then bacon and cake are used.
3. If donuts are not used, apricots are not used.
4. Apricots and eggs are used.

Pick **ONE** option

- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

Clear Selection

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22. Chef's Puzzle 1

Out of the nine ingredients, a chef wants to use exactly five. Given the following statements, how many of the 5 ingredients are known?
The ingredients are apricots, bacon, cake, donuts, eggs, figs, grapes, hazelnuts, and ice cream.

1. If bacon is not used, then grapes, hazelnuts, and ice cream are not used.
2. If apricots are used then bacon and cake are used.
3. If donuts are not used, apricots are not used.
4. Apricots and eggs are used.

Pick **ONE** option

☐ 2

☐ 3

☐ 4

☐ 5

25. Favorite Color

There are five people: Jack, Matthew, Albert, Peter, and Sebastian.
Each plays one sport out of football, cricket, volleyball, badminton, and squash.
They are from South Korea, the UK, India, China, and Russia, though not necessarily in that order.
Their favorite colors are brown, green, red, black, and yellow, again not necessarily in that order.

The following information is known-

1. Jack's favorite color is red. He is not from India or China.
2. Albert plays football. His favorite color is not yellow, and he is not from the UK.
3. Peter and Sebastian have yellow and green as their favorite colors, not necessarily in that order.
4. Matthew plays badminton. He is from Russia.
5. Peter and Sebastian are from China and UK, not necessarily in that order.
6. Brown is the favorite color of the person from India.
7. Green is the favorite color of the person from China.
8. The person from South Korea plays squash and the person from the UK plays volleyball. One of them has yellow

Whose favorite color is black?

Pick **ONE** option

☐ Matthew

☐ Albert

☐ Peter

☐ Sebastian

Clear Selection

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29. I don't think we should hire him. He seems

I don't think we should hire him. He seems like _____

Pick **ONE** option

☐ A fish out of water

☐ The black sheep of the family

☒ A wolf in sheep's clothing

☐ The apple of my eye

Clear Selection

 Saved!

30. I've been studying for this exam for months, so I

33. Logic - Sports Enthusiasts

In a company, 100 employees are randomly selected. 50 like watching football, 40 like watching hockey and the rest like both. Find out the number of employees who like to watch at least one of the games.

Pick ONE option

☐ 40

☐ 90

☐ 80

☐ 20

Selection

eger Division

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In a group of students, 60% like football, 40% like cricket. What percentage of students like only cricket if 12% of the students like both football and cricket?

Pick **ONE** option

☒ 28

☐ 14

☐ 56

☐ 42

Clear Selection

☒ Saved!

Continue

Pick ONE option

- ☐ create table orders partition by range columns (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2);
- ☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2) primary key (order_id);
- ☒ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2);
- ☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2) partition by range (order_id);

Clear Selection

☒ Saved!

9. Which of the following is a key-value store NoSQL database system?

Of the following SQL statements would correctly create a partitioned table based on the order_date column, partitioned by range?

Consider a database table "orders" with columns: order_id (Primary Key), customer_id, order_date, and total_amount. Which of the following SQL statements would correctly create a partitioned table based on the order_date column, partitioned by range?

Pick **ONE** option

- ☐ create table orders partition by range columns (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2);
- ☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2) primary key (order_id);
- ☒ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2);
- ☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2) partition by range (order_id);

Clear Selection

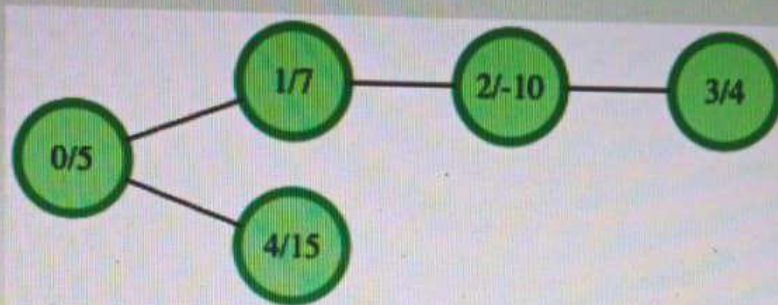


ALL



1. Best Sum Any Tree Path

Given a tree rooted at node 0 and a *value* assigned to each node, determine the maximum sum of the values along any path in the tree. The path must not be empty, and in some cases it might not go through the root. The following tree (labeled *node number/value*) is analyzed below.



Two of the possible paths are $4 \rightarrow 0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ which has a sum of $15 + 5 + 7 + -10 + 4 = 21$ and $1 \rightarrow 2 \rightarrow 3$ with a sum of $7 + -10 + 4 = 1$. A third possible path, the maximum sum path, is $4 \rightarrow 0 \rightarrow 1$ with a sum of $15 + 5 + 7 = 27$.

Function Description

Complete the function `bestSumAnyTreePath` in the editor below. It must return an integer that represents the largest value sum on any non-empty path in the tree.

`bestSumAnyTreePath` has the following parameter(s):

- `parent int[]`: Integer array where `parent[i] = j` means that node `j` is a parent of node `i`. `parent[0]` will be set to `-1` to indicate that node `0` is the root of the tree.
- `values int[]`: Integer array where `values[i]` denotes the value of node `i`.



ALL



- *parent* *int*[]): integer array where *parent*[*i*] = *j* means that node *j* is a parent of node *i*. *parent*[0] will be set to -1 to indicate that node 0 is the root of the tree.
- *values* *int*[]): integer array where *values*[*i*] denotes the value of node *i*

Constraints

- $1 \leq n \leq 10^5$
- *parent*[0] = -1
- $0 \leq \text{parent}[i] < n$ for $1 \leq i < n$
- $-1000 \leq \text{values}[i] \leq 1000$
- the *parent* array defines a valid tree

▼ Input Format Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer *n*, the size of arrays *parent* and *values*.

Each of the following *n* lines contains an integer, *parent*[*i*].

The next line, contains the integer *n*

Each of the following *n* lines contains an integer, *values*[*i*].

▼ Sample Case 0

Sample Input

5

-1

0

1

2

0

5

-2

1h 38m
left

The next line, contains the integer n
Each of the following n lines contains an integer,
 $values[i]$.

▼ Sample Case 0

Sample Input

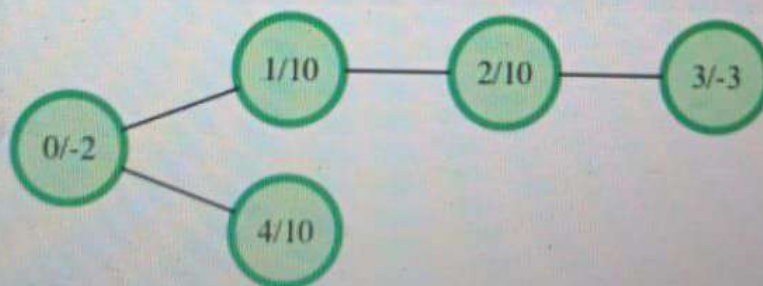
```
5
-1
0
1
2
0
5
-2
10
10
-3
10
```

Sample Output

28

Explanation

The tree looks like:



The path with the largest sum of values starts at node 2
and ends at node 4. The sum of node values is $10+10+(-2)+10 = 28$.

▼ Sample Case 0

Sample Input

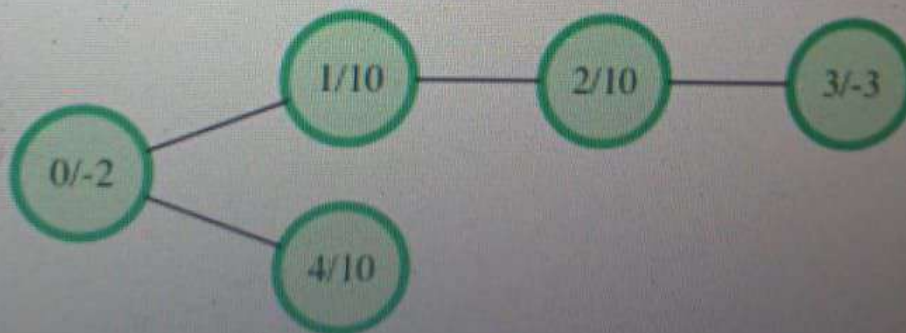
5
-1
0
1
2
0
5
-2
10
10
-3
10

Sample Output

28

Explanation

The tree looks like:



The path with the largest sum of values starts at node 2 and ends at node 4. The sum of node values is $10 + 10 + (-2) + 10 = 28$.